

Question	Answer	Marks
10(a)	ultrasound production: vibrating quartz crystal	B1
	X-ray production: electrons hitting metal target	B1
	ultrasound detected wave: reflected	B1
	X-ray detected wave: transmitted	B1
10(b)(i)	$I = I_0 \exp(-\mu x)$	C1
	$\ln(0.72) = -6.2\mu$ $\mu = 0.053 \text{ cm}^{-1}$	A1
10(b)(ii)	$I / I_0 = \exp(-9.3 \times 0.053)$ $(= 0.61)$	C1
	percentage attenuated $= 100 \times (1.00 - 0.61)$ $= 39\%$	A1